

Complete Guide to Dice Reasoning

Comprehensive Theory, 12 Solved Examples, 25 Practice Questions & Answer Key

1. Concept & Theoretical Rules

A cube has 6 faces, 8 corners (vertices), and 12 edges. In competitive reasoning exams, questions on dice test your spatial visualization and logical deduction abilities.

1.1 Types of Dice

Standard Dice (Ordinary Dice)

In a standard dice, the sum of numbers on any two opposite faces is **always equal to 7**.

- Opposite of 1 is 6 ($1 + 6 = 7$)
- Opposite of 2 is 5 ($2 + 5 = 7$)
- Opposite of 3 is 4 ($3 + 4 = 7$)

Note: Adjacent faces NEVER sum to 7 in a standard dice.

Ordinary / Non-Standard Dice

In an ordinary dice, the sum of numbers on adjacent faces can be equal to 7. The opposite faces do not strictly sum to 7.

1.2 Fundamental Rules for Solving Dice Problems

Rule 1: Adjacency Rule

Two opposite faces can **never** be adjacent to each other, and two adjacent faces can **never** be opposite to each other.

Rule 2: Two Common Numbers / Symbols Rule

If two positions of the same dice show **two common numbers/symbols**, then the remaining third un-common faces are guaranteed to be **opposite to each other**.

*Example: Position A shows (2, 3, 5) and Position B shows (2, 3, 6). Here 2 & 3 are common, so **5 is opposite to 6**.*

Rule 3: One Common Number Rule (Common Rotation Rule)

If two positions show only **one common face**, write down the numbers of both dice starting from the common number moving in a **clockwise (or counter-clockwise) direction**. The numbers that align in the sequence are opposite to each other.

Example: Position A = (3, 1, 4) & Position B = (3, 2, 5). Common is 3.

Clockwise from 3: A $\rightarrow$ 3 - 1 - 4 | B $\rightarrow$ 3 - 2 - 5.

*Therefore: **1 is opposite to 2, 4 is opposite to 5, and 3 is opposite to 6** (the remaining face).*

Rule 4: Open Dice (Unfolded Cube) Rules

When a cube is unfolded into a 2D net, alternate faces (separated by one square) are opposite to each other.

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[ 1 ]
[ 2 ] [ 3 ] [ 4 ] [ 5 ]
[ 6 ]
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In this standard cross net layout:

- Face 2 is opposite to Face 4 (alternate in row)
- Face 3 is opposite to Face 5 (alternate in row)
- Face 1 is opposite to Face 6 (top and bottom flaps)

2. Solved Examples (12 Detailed Problems)

EXAMPLE 1

Question: What number is opposite to 3 on a standard dice?

Solution:

On a standard dice, the sum of opposite faces is always 7. Therefore, Opposite of $3 = 7 - 3 = 4$.

EXAMPLE 2

Question: Two positions of a dice are given: Position 1 shows (1, 2, 5) and Position 2 shows (1, 3, 6). Which number is opposite to 2?

Solution:

Common number is 1. Moving clockwise from 1:

Position 1: $1 \rightarrow 2 \rightarrow 5$

Position 2: $1 \rightarrow 3 \rightarrow 6$

Corresponding pairs are (2, 3) and (5, 6). Thus, opposite of 2 is **3**.

EXAMPLE 3

Question: Two positions of a dice show (4, 2, 6) and (4, 2, 1). Which face is opposite to 6?

Solution:

By Rule 2 (Two Common Numbers): Numbers 4 and 2 are common in both positions. Therefore, the remaining third faces must be opposite to each other. Hence, 6 is opposite to **1**.

EXAMPLE 4

Question: Three positions of a dice are shown: (A, B, C), (B, D, E), and (C, E, F). What letter is opposite to B?

Solution:

Comparing Position 1 (A, B, C) and Position 2 (B, D, E): Common element is B.

Clockwise from B: Position 1 → B-C-A, Position 2 → B-E-D.

So, C is opposite to E, and A is opposite to D. The remaining face opposite to B must be **F**.

EXAMPLE 5

Question: An open dice net has vertical column: [A], [B], [C], [D] with side flaps [E] attached to B and [F] attached to C. Which face is opposite to B?

Solution:

In a straight vertical line of alternate faces: A is opposite to C, B is opposite to **D**, and E is opposite to F.

EXAMPLE 6

Question: If numbers 1, 2, 3, 4, 5, 6 are written on a dice and adjacent numbers to 4 are 1, 2, 5, 6. Which number is opposite to 4?

Solution:

Since 1, 2, 5, and 6 are all adjacent to 4, the only remaining number that is NOT adjacent must be opposite to 4. Therefore, opposite to 4 is **3**.

EXAMPLE 7

Question: Two positions show symbols: (★, ●, ▲) and (★, ■, ◆). Which symbol is opposite to ●?

Solution:

Common symbol is ★. Moving clockwise from ★:

Pos 1: ★ → ● → ▲

Pos 2: ★ → ■ → ◆

Thus, ● is opposite to **■ (Square)**.

EXAMPLE 8

Question: Two positions of a dice show (3, 5, 2) and (3, 5, 6). What is opposite to 3?

Solution:

Here 3 and 5 are common, so 2 is opposite to 6. Faces adjacent to 3 are 5, 2, and 6. The remaining hidden faces are 1 and 4. Thus, opposite to 3 can be either **1 or 4**.

EXAMPLE 9

Question: In a dice, faces are colored Red, Blue, Green, Yellow, Orange, White. Red is opposite to Green, Blue is opposite to Yellow. Which color is opposite to Orange?

Solution:

Pairs formed so far: (Red, Green) and (Blue, Yellow). The remaining pair of faces must be Orange and **White**.

EXAMPLE 10

Question: Four positions of a dice are shown. Number 1 appears on the top face of Pos 1 (1,2,3), Pos 2 (1,3,5), Pos 3 (1,5,4), Pos 4 (1,4,2). What number is at the bottom face?

Solution:

In all four positions, 1 is adjacent to 2, 3, 5, and 4. The only number not adjacent to 1 is 6. Thus, 6 is at the bottom (opposite to 1). Answer is **6**.

EXAMPLE 11

Question: In an open net layout: Row 1 = [1, 2, 3, 4], Flap top = [5] above 2, Flap bottom = [6] below 2. What face is opposite to 5?

Solution:

In the horizontal row: 1 is opposite to 3, 2 is opposite to 4. The top and bottom flaps are opposite to each other. So, 5 is opposite to **6**.

EXAMPLE 12

Question: Two positions show (6, 1, 4) and (5, 3, 2). If 6 is opposite to 5, which face is opposite to 1?

Solution:

Given 6 is opposite to 5. Aligning clockwise around 6 and 5 respectively: Pos 1 clockwise (6 → 1 → 4) corresponds to Pos 2 counter-clockwise (5 → 3 → 2). Thus 1 is opposite to **3** (or 2 depending on orientation; here 1 is opposite to 3).

3. Practice Questions (25 MCQ Questions)

Q1. On a standard dice, what is the sum of numbers on the bottom face and top face?

- | | |
|-------|-------|
| (A) 5 | (B) 6 |
| (C) 7 | (D) 8 |

Q2. Two positions of a dice show faces (2, 4, 3) and (2, 5, 6). What number is opposite to 4?

- | | |
|-------|-------|
| (A) 5 | (B) 6 |
| (C) 1 | (D) 3 |

Q3. Two positions show (1, 6, 5) and (1, 6, 4). Which number is opposite to 5?

- | | |
|-------|-------|
| (A) 2 | (B) 3 |
| (C) 4 | (D) 6 |

Q4. On a standard dice, if 5 is on the front face, what number is on the back face?

- | | | | |
|-------|-------|-------|-------|
| (A) 1 | (B) 2 | (C) 3 | (D) 4 |
|-------|-------|-------|-------|

Q5. Two positions of a dice show (3, 1, 2) and (3, 5, 4). What number is opposite to 3?

- | | | | |
|-------|-------|-------|-------|
| (A) 6 | (B) 4 | (C) 2 | (D) 5 |
|-------|-------|-------|-------|

Q6. If faces of a cube are labeled A, B, C, D, E, F. Positions show (A, B, C) and (A, D, E). What is opposite to A?

- | | | | |
|-------|-------|-------|-------|
| (A) B | (B) C | (C) F | (D) E |
|-------|-------|-------|-------|

Q7. In an open net horizontal chain [1][2][3][4], face 1 is opposite to which face?

- | | | | |
|-------|-------|-------|-------|
| (A) 2 | (B) 3 | (C) 4 | (D) 5 |
|-------|-------|-------|-------|

Q8. Two positions of a dice show (5, 2, 1) and (5, 2, 6). Which face is opposite to 1?

- | | | | |
|-------|-------|-------|-------|
| (A) 3 | (B) 4 | (C) 6 | (D) 5 |
|-------|-------|-------|-------|

Q9. Three positions show (1, 2, 4), (2, 4, 5), and (3, 4, 5). Which number is opposite to 2?

- | | | | |
|-------|-------|-------|-------|
| (A) 1 | (B) 3 | (C) 5 | (D) 6 |
|-------|-------|-------|-------|

Q10. On a standard dice, what is the product of opposite faces 2 and 5?

(A) 7

(B) 10

(C) 12
(D) 14

Q11. Two positions show symbols (+, -, *) and (+, /, ?). Which symbol is opposite to - ?

(A) /

(B) ?

(C) +
(D) *

Q12. If adjacent faces of face 6 are 1, 2, 3, and 4, which number is opposite to 6?

(A) 5

(B) 1

(C) 2
(D) 3

Q13. Two positions of a dice show (6, 3, 2) and (6, 5, 1). What number is opposite to 3?

(A) 1

(B) 2

(C) 5
(D) 4

Q14. An open cube net has vertical strip [X, Y, Z, W]. Which face is opposite to Y?

(A) X

(B) Z

(C) W
(D) Cannot be determined

Q15. If two dice positions show (2, 3, 6) and (4, 1, 5), and 2 is opposite to 4, then 3 is opposite to?

(A) 1

(B) 5

(C) 6
(D) Either 1 or 5

Q16. On a standard dice, what is the face opposite to 6?

(A) 1

(B) 2

(C) 3
(D) 5

Q17. Two positions show (4, 1, 5) and (4, 3, 2). What face is opposite to 5?

(A) 1

(B) 2

(C) 3
(D) 6

Q18. Two positions show (3, 4, 1) and (3, 4, 2). What face is opposite to 1?

(A) 2

(B) 5

(C) (D)
6 3

Q19. Which of the following CANNOT be adjacent faces in a standard dice?

(A) (1, 2)

(B) (3, 4)

(C) (D)
(2, (5,
3) 6)

Q20. Position 1 shows (6, 5, 4) and Position 2 shows (6, 2, 1). What number is opposite to 6?

(A) 3

(B) 1

(C) (D)
2 5

Q21. Two positions show colors (Red, Yellow, Blue) and (Red, Green, White). What color is opposite to Yellow?

(A) Green

(B) White

(C) (D)
Red Blue

Q22. In an open net, faces are arranged: Top [1], Middle strip [2, 3, 4, 5], Bottom [6]. Opposite of 3 is?

(A) 2

(B) 4

(C) (D)
5 6

Q23. If 1 is opposite to 6, 2 is opposite to 5, which face is adjacent to both 3 and 4?

(A) 1

(B) 2

(C)
All
other (D)
faces None
(1, 2,
5, 6)

Q24. Two positions show (5, 3, 6) and (5, 4, 1). Opposite of 3 is?

(A) 4

(B) 1

(C) (D)
2 6

Q25. How many total vertices (corners) does a standard dice cube have?

(A) 6

(B) 8

(C) (D)
12 16

4. Answer Key

Below is the complete answer key for all 25 practice questions:

Q.No	Ans	Explanation / Key Concept	Q.No	Ans	Explanation / Key Concept
Q1	(C) 7	Standard dice sum = 7	Q14	(C) W	Alternate elements Y & W
Q2	(A) 5	Clockwise from 2: 4→5	Q15	(A) 1	Orientation rule matching
Q3	(C) 4	Two common faces (1, 6)	Q16	(A) 1	Standard dice: 7 - 6 = 1
Q4	(B) 2	Standard dice: 7 - 5 = 2	Q17	(B) 2	Clockwise from 4: 5→2
Q5	(A) 6	Clockwise from 3; 6 remaining	Q18	(A) 2	Two common faces (3, 4)
Q6	(C) F	Remaining letter is F	Q19	(B) (3, 4)	Sum is 7 (Opposite, not adjacent)
Q7	(B) 3	Alternate face in net	Q20	(A) 3	Common 6; 3 is remaining face
Q8	(C) 6	Two common faces (5, 2)	Q21	(A) Green	Clockwise from Red: Yellow→Green
Q9	(B) 3	Common faces 4 & 5 gives 2 opposite 3	Q22	(C) 5	Alternate in strip [2, 3, 4, 5] → 3 & 5
Q10	(B) 10	$2 \times 5 = 10$	Q23	(C) All	3 & 4 are opposite, so all 4 faces adjacent
Q11	(A) /	Clockwise from +	Q24	(A) 4	Clockwise from 5: 3→4
Q12	(A) 5	Non-adjacent remaining face	Q25	(B) 8	A cube has 8 vertices/corners
Q13	(C) 5	Clockwise from 6: 3→5			